

Project Tricorder Mk I · Field manual · Rev 0.6.78

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Companion PDF: Project-Tricorder-MkI-Field-Manual-0.6.78.pdf.

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Face

I Inquiry · **II** Sensor · **III** Sentry (continuous default) · **IV** Audio (off → green listen → blue planes book → yellow alerts book → off) · **T** / **XLATE** Translator · **H** Horizon · **S** Sat/Stars · **O** Sol (space → earth weather → moon phase → sunrise/sunset → moonrise/moonset → off) · **L E V P** logs · **IPL** / **ALL** abort · **STT** kill speech.

ROBO four-state cycle: dark = Kokoro · green ROBO = Android robot voice · yellow [**H/F R**] = robot + hands-free top-level listen · red [**MUTE**] = silent observer (CRT and log only, no voice, no alert tones). Next touch returns dark / Kokoro.

Spoken overlays: **SETTINGS**, **BACK END** / **BACKEND** / **CAMERA**, **HELP**, **DIAGNOSTICS**, **COMPUTER**, **CLOCK**, **DICTATION**, and the **TRICORDER ...** menu.

0.6.77–0.6.78 — Audio books and heap hygiene

IV cycle (0.6.77). First press arms Audio (green, YAMNet running). Second press keeps listen armed and opens the *planes book* (blue): last 80 overhead cards with PREV/NEXT, no wait for the next aircraft. Third press opens the *alerts book* (yellow): civic / siren cards. Fourth press is off. New planes do not steal the alerts book and the other way around. Empty book: “0 on file; next event will land here.” **DISMISS** inside a book leaves the book on screen.

NAV callsign (0.6.77). The moving green dot is the Comlink name (Jim), not CAMP. If you are standing on camp, CAMP sits above the dot and the name sits below it. Other linked tubes stay cyan with their names. Save the name in Comlink if the map still says YOU.

Heap (0.6.78). A silent bounce with no crash dialog is Android reclaiming RAM after a long Audio + flights day. Thumbs recycle after decode and cap at 320 px. The flight book on disk keeps photo URLs, not 80 in-RAM JPEGs. Field-log summary streams the last 1800 rows. Auto-backup runs at most every five minutes. GPS 1.5 s / 2 m. **largeHeap** plus **onTrimMemory** / **onLowMemory** drop old card photos. The system can still kill any all-day listener if another app needs RAM; we just stop volunteering first.

0.6.72–0.6.76 — sun, luna, mesh pins, airframe book

Airframe carousel (0.6.72). PREV/NEXT holds the last 80 unique overhead *cards* (photo + facts). The pointer is clamped after the cap, so the face never reads 81/80. Every overhead still writes a field-log row. **SUMMARIZE READINGS** and log review walk the full book, not the 80.

SOL sunrise / sunset (0.6.73–0.6.74). Fourth press of O (after moon phase). NOAA / Wikipedia sunrise equation on the last GPS fix. Longitude is converted west-positive; the day is anchored on *local noon* so times print in the phone’s zone (e.g. 6:47 AM EDT), not UTC. Polar day/night if the math says so. Voice **SUN**, **SUNRISE**, **SUNSET**.

Mesh GPS on NAV (0.6.73 / 0.6.77). Linked tricorders beacon lat/lon every 60 s over Nearby and LoRa (and on hello). NAV map plots each unit at its last fix with the Comlink callsign. Your tube is the green moving name. Peers are cyan; dull after three minutes with no fix. Both units need 0.6.73+ and a GPS lock. Comlink must be running.

SOL moonrise / moonset (0.6.75–0.6.76). Fifth press of O, label **luna**. Meeus-lite moon position, local-day horizon scan with parallax. No Skyfield, no DE421 file. Face stamps **approx ± 3 min not USNO**. Rise and set include compass heading (for 12550 on 25 Sep 2026 that is about 91° E / 264° W). Small rose marks R and S. Voice **MOONRISE**, **MOONSET**, **LUNA**. If the moon does not cross the horizon that civil day the panel says stays up / stays down.

0.6.67–0.6.71 — cover cockpit

Designed for the Galaxy Fold cover glass (unit folded shut). Outer display only. Inner unfolded screen is left alone.

Arm Horizon first (H on). Rotate the closed unit to landscape. The face hides. The cover becomes an edge-to-edge artificial horizon: sky / ground, aircraft symbol, G tape, heading rose, and the instrument stack. Rotate back to portrait and the tricorder face returns at normal size.

If H is off, landscape stays the ordinary (narrow) layout. Inquiry, L/E/V/P log keys, and the LED grille are hidden in cockpit so they do not eat the ADI.

Landscape HUD only (portrait strip unchanged): HDG, SPD, BARO, ALT, GPS ALT each on its own line, large phosphor digits. Heading rose is a large N/E/S/W card with ticks and a thick needle. Portrait rose stays the small 14 px ring.

0.6.68–0.6.69: cover landscape often reports orientation angle 0 inside the WebView. The old remap put screen-Y negative and pegged P/R at $\pm 90^\circ$. Cockpit now maps device X onto screen-Y and keeps that axis positive. Portrait gravity math is untouched.

0.6.62–0.6.66 — persistence, API health, LoRa

Field book and the overhead queue are written after each detection and reloaded on launch. Do not treat RAM as the book.

OpenSky default poll is five minutes. HTTP 429 backs off twenty minutes. Robot voice (MUTE honored): `warning: opensky is 429-ing, rate limit for your IP`. Same pattern for other watch APIs that return a hard status.

DIAGNOSTICS now prints an API check block on the CRT: CityAlert / civic, NOAA / IPAWS, USGS, OpenSky. Status, milliseconds, hit count.

Indoor flights use the last GPS fix as the geobox center so Wi-Fi-only rooms still see overheads.

LoRa / Meshtastic is a second Comlink bearer, not a replacement for Nearby. SenseCAP T1000-E (or any Meshtastic node) pairs over BLE. Settings live in Comlink: radio on/off, pairing PIN (factory 123456), region, destination (broadcast 0xFFFFFFFF), hop count, filter. Nearby stays short-range; LoRa is the long-haul short-text path. Event passing is automatic once the radio is on and linked. Voice still **TRICORDER COMLINK**.

0.6.58–0.6.61 — audio look-back and printed cards

Audio false positives on a highway / pond / walk: look back a few listens. Footsteps then “horse” is demoted. “Ship traffic” next to road or pump noise is demoted. “Air traffic” from YAMNet (not the OpenSky card) next to cars is demoted. Those three plus unsure gunfire no longer get a spoken sentence. Beep-beep-beep, red triangle on the CRT, line reads `possible ship traffic / possible gunfire / possible air traffic`. Real jets over Stewart still hit OpenSky as flights. That path is unchanged.

NASA-style 3×5 field cards (print/laminate) ship as Project-Tricorder-MkI-Field-Cards-3x5.pdf. Branding strap: PROJECT TRICORDER MK. I · Ames · Grok · Claude · USA · GPL3.

0.6.55–0.6.57 — AUTOPLAY, units, backup

AUTOPLAY on an overhead card. Touch once: photo + spoken line, then airframe facts, then origin city (up to three Wikimedia Commons airport views), then dest city, then the next craft newest-first. Touch again to stop. Needs Earth link for photos and dossiers.

Spoken line is built from numbers, not a regex: `tail ID N2GYH overhead United Airlines at 10000 feet 570 miles per hour mach 0.75 from San Francisco to New York City`. Settings → Aircraft English units writes native prefs. English (ft, mph) is the default.

BACKUP TRICORDER glows green and writes app-private JSON plus a Downloads sidecar. **RESTORE TRICORDER** glows yellow. 0.6.57 restores the *internal* copy first — a Downloads stub on scoped storage was why older restores said unreadable. Callsign is reapplied from the face blob.

SAVE NAME on Comlink is a real click handler (the old listener ran before the sheet existed). Name persists across leaves.

0.6.56: DISMISS AIRFRAME / DISMISS EVENT releases both CRT holds. A leftover civic hold was freezing the tube on “Airframe card dismissed” and blocking audio graphs.

0.6.52–0.6.54 — civic cards and geobox

SpotCrime keys are bound to a static IP. A phone cannot use them. Civic watch uses **CityAlert** public GeoJSON (no key, any IP), plus IPAWS CAP and USGS quakes. Coverage is 511 / open data / hazards / community reports — not Citizen Inc. crime pins.

Civic hits stack like airframes: type icon (road, crime, fire, quake, weather, medical, AMBER, civil), **HEAR ALERT AGAIN**, PREV/NEXT, DISMISS EVENT. All civic/safety speech is robot voice. IPAWS/quakes/CityAlert never go through Kokoro.

Settings sliders (half-width km around last GPS fix): Flight box 8–80 default 28 · Civic/511 box 8–80 default 31 · Quake radius 50–800 default 400. Stored in prefs and the backup blob.

OPEN LOCAL SCANNER and OPEN PULSEPOINT still hand off to the official sites. We do not wrap their audio.

0.6.50–0.6.51 — civic voice and Comlink persist

Local civic lines were Kokoro. They are robot-only. Armed notices are short. Comlink callsign writes SharedPreferences with commit plus a sidecar file.

0.6.46–0.6.47 — Comlink

Automatic Nearby P2P_CLUSTER mesh in the APK. GUID is TC- plus a hash of ANDROID_ID (not IMEI). Callsign is assigned in the Comlink pane. Yellow = heard, green = linked. Touch a name to mute that unit's alerts. PTT: speak; far tube says **Beep beep beep**. From NAME. ... in robot voice. Alerts arrive as **Remote tricorder Spock detected** Voice: TRICORDER COMLINK, TRICORDER COM LINK, TRICORDER COMMLINK, or tap Comlink on the nameplate. Both units need 0.6.46+.

0.6.45 — scrollable CRT

The phosphor pane is a fixed-height tube. In NAV it scrolls as one window: matrix, horizon, map, GO/MARK/CAMP/BACKTRACK. Swipe on the tube. NAV also raises the pane to 300 px.

0.6.44 — matrix under STANDBY

CAMP / YOU / DEST lat-lon and SAID sit at the **top** of the CRT under STANDBY. Anything drawn under the map is clipped by the 230 px bezel.

0.6.42–0.6.43

0.6.42 declared `const strip` twice and blanked the face. 0.6.43 is the repair. Do not sideload 0.6.42.

0.6.39–0.6.41 — dest parse and live fix

Hemispheres: 41.50N 74.01W, N 41.50 W 74.01, 41.50, -74.01. Bare positive longitude is east. Speech that drops the decimal (41 50 N 74 01 W) is rebuilt. NAV pulls GPS from the native bridge every paint so YOU/CAMP appear without waiting for a new location callback. TRICORDER CAMP marks camp here. TRICORDER BACKTRACK points dest at camp.

0.6.38 — walking NAV (Horizon second press)

H cycles: off → horizon → **NAV** → off. NAV is line-of-sight walking only. No roads, no driving, not for flight. Set dest as lat, lon, plus code, or ///word.word.word (W3W needs the settings key). GO computes bearing and distance. MARK drops a waypoint. CAMP stores this fix as home. BACKTRACK sets dest to camp. North-up symbol map on the CRT under the horizon: YOU, trail, DEST, CAMP, waypoints. Off-track chime if you wander more than 30 m from the line. GPS stale > 8 s: dead-reckon along heading at a walk. Voice: TRICORDER NAV, TRICORDER CAMP, TRICORDER MARK, TRICORDER BACKTRACK, TRICORDER NAVIGATE TO 41.50N 74.01W.

0.6.37 — CRT stack and cattle

Detector bars on Audio / Sensor / Sentry now stack **weakest on top, HIGHLY LIKELY on the bottom** of the CRT so the green winner stays in frame after auto-scroll. Wrapped labels no longer hide the lead bar.

Bucket bug: the linker used `\bcat`, which matched YAMNet's *Cattle*, *bovinae*. Mooing therefore painted as CAT. Cattle / moo / cow is its own bucket; cat only matches `cat` / `cats`.

0.6.36 — flight readings isolation

By-airframe / by-tail timelines were walking neighboring YAMNet lines (Silence, Speech) because every listen wrote a field-log row and the bar used file-line index.

Fix: a flights bar only accepts `kind=fight` (or a flight/context row with tail/type). Playback is by log **id**, not “next line.” Raw listen noise (Silence, Speech, Quiet) stays in the audio log and is not a field-log event unless BirdNET named a bird.

Old mixed rows remain in the book. New overheads chart clean.

0.6.34–0.6.35 — live detector and LED meter

TOP 3. Audio, Sensor, and Sentry keep the raw percent list. Scroll the CRT *up* for it. Under that list a fused TOP 3: same family collapses (barred owl + hoot → owl). #1 HIGHLY LIKELY glows green when it leads; other possibles yellow; junk tails red. Charts refresh every listen/scan. Auto-scroll parks the phosphor on the bars.

LED. Scroll text is #1 LABEL 72%. Right-hand vertical tube: 0% bottom, 100% top. Fill is solid below the reading, outline only above it. Amber pointer on the live percent. Heading/ACC chatter does not overwrite that while a detection is live.

Readings drill. Top chart is still HUMANS / VEHICLES / FLIGHTS / Touch a bar for that category's sub-bars and its log timeline. BACK returns to the summary. Flights: **By tail** and **By airframe**. Photo on a flight row opens the airframe card. FROM/TO open the city dossier.

Overhead set. A pinned flight card also shows a small queue chart of tails and types already in the overhead set (up to several concurrent hits).

0.6.33 — MUTE

Fourth ROBO state. Red [MUTE]: no Kokoro, no robot voice, no klaxon/chime. Screen and field log still write. Silent observer / camp unit that must not announce itself. Next touch returns Kokoro (dark).

0.6.32 — hands-free TRICORDER menu

Yellow [H/F R]: say the ship name first, then the function. Leave a mode with the face button.

Commands: TRICORDER SENSOR · AUDIO · SENTRY · TRANSLATOR / TRANSLATE · HORIZON · SATELLITES · SOL · CLOCK · DICTATION · SETTINGS · BACK END · DIAGNOSTICS · SUMMARIZE READINGS · HELP.

While yellow, speech that is not a TRICORDER line is ignored. Audio, translator, clock, and dictation own the microphone while they are up. Aliases include “trick order” and “tric order.”

0.6.30–0.6.31 — satellite clock, moon, places

CLOCK. Say CLOCK or TRICORDER CLOCK. Analog or digital, 12- or 24-hour. After the face opens: analog, digital, twelve hour, twenty four hour, timer coffee five minutes, cancel timer coffee, cancel all, explain, terminate.

GNSS lock uses Android GnsClock (GPS TimeNanos minus full bias minus bias, leap seconds to UTC). Display 4 Hz; hardware epoch ~1 Hz interpolated. Fold 3 open-sky uncertainty is typically tens to a few hundred nanoseconds — field clock, not a standard. Until lock, device wall time. Field-log timestamps stay ordinary milliseconds.

Moon / sun / luna. Sol cycle: space weather + orbits + HF/VHF/UHF → local earth forecast (six hours + three-day outlook, scroll) → moon phase → sunrise/sunset → moonrise/moonset → off. Voice MOON / MOON PHASE, SUN / SUNRISE, LUNA / MOONRISE / MOONSET.

Places. Logs, sensed events, and the horizon carry city/state when Geocoder has it, plus an offline Open Location Code (plus code). Optional what3words key in settings. Raw lat/lon still stored.

0.6.28 — MEDICAL extracts, pinned airframe, city BACK

MEDICAL. WikiMed is a title catalog. Body and photographs from English Wikipedia. Say medical pink eye.

Pinned airframe. Overhead card stays until DISMISS AIRFRAME. Audio may still speak alerts; it does not steal the picture.

City BACK. FROM / TO opens the city dossier. BACK TO AIRFRAME returns the same craft photo.

0.6.26–0.6.27 — flights + knowledge banks

Overhead card. OpenSky + adsbdb type/owner/tail. Photo: Planespotters, airport-data, JetPhotos. JPEG in the field log. Touch photo for airframe facts. Touch FROM/TO for city dossier.

One-shot: memory alpha Airbus A320. Bare bank name opens the prompt and arms the mic. TERMINATE closes. MEMORY ALPHA is Simple English Wikipedia, cached on device. No 30 MB illustrated ZIM is shipped.

0.6.22–0.6.25 — readings

SUMMARIZE READINGS [vehicles|flights|birds|...] [day|week|month|all]. Letter bars A B C, QTY, color legend. Touch a bar for that category's timeline. Day/Week/Month/All chips live.

TTS length slider 400–8000. Unit does not classify its own speech as humans detected while talking.

Dictation (0.6.15–0.6.21)

Wake: `dictation` or `TRICORDER DICTATION. NEW · EDIT nnnn · LOG nnnn (CONFIRMED) · SUMMARIZE nnnn · MINDMAP keyword · SPEAK nnnn · KILL nnnn (CONFIRMED) · TERMINATE. ZULU` then `SAVE` or `ABORT`.

Buffer: `CURSOR MINUS / PLUS word · BACKSPACE · NEWLINE · CUT / CUT WORD · PASTE · HOME / END. PUNCTUATION PERIOD` (also bare `PERIOD / DOT`).

Watch / IPL (0.6.9–0.6.14)

Sentry captions every dwell. Sure weapon: klaxon + `WEAPON DETECTED x3`. Possible weapon: chime. Motion does not gate.

Audio priority: gunfire x3 + klaxon; scream; siren x3; air/rail/ship; vehicles; named animal; humans x3; entertainment. Fireworks can look like gunfire.

IPL aborts in-flight work. Kokoro slider kills speech. Text over 80 characters is chunked.

SOL / sky / horizon

O Sol cycles space + HF/VHF/UHF, earth forecast, moon phase, sunrise/sunset, moonrise/moonset, then off. S toggles satellites and stars. Portrait horizon is the 96 px strip: heading, G, GPS speed, baro altitude proxy, place line. Closed-fold landscape + H on is cover cockpit (see 0.6.67–0.6.71). NAV second press of H plots YOU, CAMP, DEST, and named Comlink units. All of it is a toy. Not for flight.

Backends

Termux four-pack on loopback: chat :8080, vision :8081, image :8188, tts :5000. Launch with `~/bin/tricorder start`. Image sliders: steps and size (256 / 384 / 512). See `TERMUX-BACKEND-INSTALL.md` in the source tree.

On-device packs (APK, no Termux): YAMNet, optional BirdNET fetch, flora/fauna fetch, translator language packs. First-boot will pull missing packs when Earth link exists.